INCH-POUND

MIL-PRF-39006/21A
17 April 2002
SUPERSEDING
MIL-C-39006/21
25 January 1972

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM,
(POLARIZED, SINTERED SLUG) (EXTENDED RANGE), 85°C (VOLTAGE DERATED TO 125°C),
ESTABLISHED RELIABILITY,
STYLE CLR69

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-39006.

INACTIVE FOR NEW DESIGN AFTER 23 JUNE 1981. FOR NEW DESIGN, USE MIL-PRF-39006/22.

NOTE: The above entry consolidates all previous specification action concerning inactivation. Failure rate level "L" was made inactive for new design on 27 May 1976. All other failure rate levels were inactivated by Notice 1 and 2 dated 9 October 1981 and 11 June 1982, respectively.

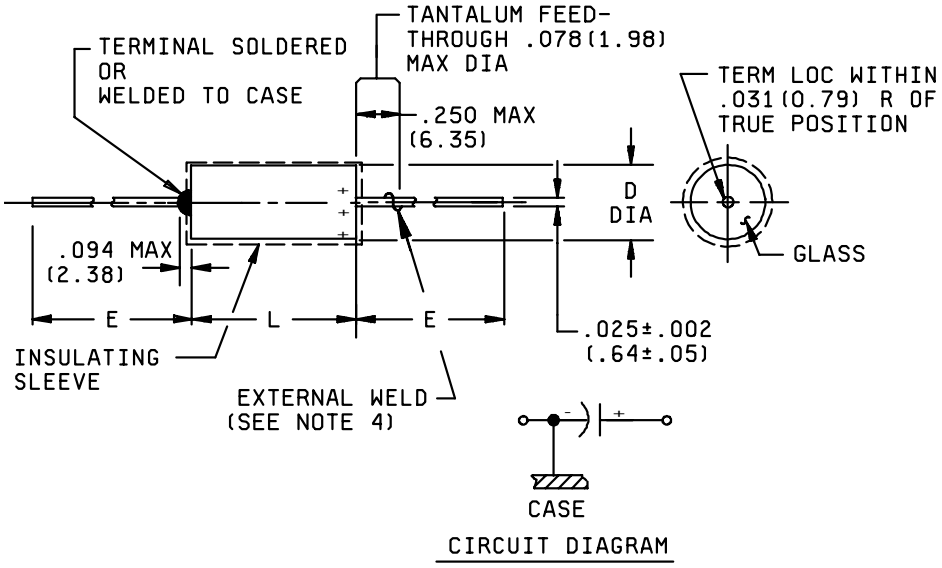


FIGURE 1. Style CLR69 capacitors.

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Case size	Dimensions			
	Basic case		Insulated case	
	L 1/ +.031 (0.79) -.016 (0.41)	D ±.016 (0.41)	D max	E ±.250 (6.35)
T1	.453 (11.51)	.188 (4.78)	.219 (5.56)	1.500 (38.10)
T2	.641 (16.28)	.281 (7.14)	.312 (7.92)	2.250 (57.15)
T3	.766 (19.46)	.375 (9.53)	.406 (10.31)	2.250 (57.15)
T4	1.062 (26.97)	.375 (9.53)	.406 (10.31)	2.250 (57.15)

1/ Length of basic case; sleeving shall be as specified in MIL-PRF-39006.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR69 capacitors - Continued.

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TABLE I. Capacitor characteristics.

Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at			Case size	PIN M39006/21- Failure rate level (%/1,000 hour)				
			+25°C	+85°C & +125°C					-55°C	+85°C	+125°C		L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
volts, dc	µF	percent	µA	µA	percent	volts, dc	volts, dc	ohms	percent	percent	percent						
6	220	±20	2	9	50	4	6.9	36	-64	+13	+16	T1	0001	0089	0177	0265	0353
6	220	±10	2	9	50	4	6.9	36	-64	+13	+16	T1	0002	0090	0178	0266	0354
6	820	±20	3	14	155	4	6.9	18	-88	+16	+20	T2	0003	0091	0179	0267	0355
6	820	±10	3	14	155	4	6.9	18	-88	+16	+20	T2	0004	0092	0180	0268	0356
6	1,500	±20	5	20	172	4	6.9	18	-90	+20	+25	T3	0005	0093	0181	0269	0357
6	1,500	±10	5	20	172	4	6.9	18	-90	+20	+25	T3	0006	0094	0182	0270	0358
6	2,200	±20	6	24	170	4	6.9	13	-90	+25	+30	T4	0007	0095	0183	0271	0359
6	2,200	±10	6	24	170	4	6.9	13	-90	+25	+30	T4	0008	0096	0184	0272	0360
8	180	±20	2	9	41	5	9.2	45	-60	+13	+16	T1	0009	0097	0185	0273	0361
8	180	±10	2	9	41	5	9.2	45	-60	+13	+16	T1	0010	0098	0186	0274	0362
8	680	±20	3	14	130	5	9.2	22	-83	+16	+20	T2	0011	0099	0187	0275	0363
8	680	±10	3	14	130	5	9.2	22	-83	+16	+20	T2	0012	0100	0188	0276	0364
8	1,500	±20	5	20	170	5	9.2	18	-90	+20	+25	T3	0013	0101	0189	0277	0365
8	1,500	±10	5	20	170	5	9.2	18	-90	+20	+25	T3	0014	0102	0190	0278	0366
8	1,800	±20	7	25	138	5	9.2	14	-90	+25	+30	T4	0015	0103	0191	0279	0367
8	1,800	±10	7	25	138	5	9.2	14	-90	+25	+30	T4	0016	0104	0192	0280	0368
10	150	±20	2	9	34	7	11.5	54	-55	+13	+16	T1	0017	0105	0193	0281	0369
10	150	±10	2	9	34	7	11.5	54	-55	+13	+16	T1	0018	0106	0194	0282	0370
10	560	±20	3	16	106	7	11.5	27	-77	+16	+20	T2	0019	0107	0195	0283	0371
10	560	±10	3	16	106	7	11.5	27	-77	+16	+20	T2	0020	0108	0196	0284	0372
10	1,200	±20	5	20	137	7	11.5	18	-88	+20	+25	T3	0021	0109	0197	0285	0373
10	1,200	±10	5	20	137	7	11.5	18	-88	+20	+25	T3	0022	0110	0198	0286	0374
10	1,500	±20	7	25	114	7	11.5	15	-88	+25	+30	T4	0023	0111	0199	0287	0375
10	1,500	±10	7	25	114	7	11.5	15	-88	+25	+30	T4	0024	0112	0200	0288	0376
15	100	±20	2	9	30	10	17.2	72	-44	+13	+16	T1	0025	0113	0201	0289	0377
15	100	±10	2	9	30	10	17.2	72	-44	+13	+16	T1	0026	0114	0202	0290	0378
15	390	±20	3	16	74	10	17.2	31	-66	+16	+20	T2	0027	0115	0203	0291	0379
15	390	±10	3	16	74	10	17.2	31	-66	+16	+20	T2	0028	0116	0204	0292	0380
15	820	±20	6	24	111	10	17.2	22	-77	+20	+25	T3	0029	0117	0205	0293	0381
15	820	±10	6	24	111	10	17.2	22	-77	+20	+25	T3	0030	0118	0206	0294	0382
15	1,000	±20	8	32	92	10	17.2	17	-77	+25	+30	T4	0031	0119	0207	0295	0383
15	1,000	±10	8	32	92	10	17.2	17	-77	+25	+30	T4	0032	0120	0208	0296	0384
25	68	±20	2	9	22	15	28.8	90	-40	+12	+15	T1	0033	0121	0209	0297	0385
25	68	±10	2	9	22	15	28.8	90	-40	+12	+15	T1	0034	0122	0210	0298	0386
25	270	±20	3	16	55	15	28.8	33	-62	+13	+16	T2	0035	0123	0211	0299	0387
25	270	±10	3	16	55	15	28.8	33	-62	+13	+16	T2	0036	0124	0212	0300	0388

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TABLE I. Capacitor characteristics - Continued.

Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at			Case size	PIN M39006/21- Failure rate level (%/1,000 hour)				
			+25°C	+85°C & +125°C					-55°C	+85°C	+125°C		L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
volts, dc	µF	percent	µA	µA	percent	volts, dc	volts, dc	ohms	percent	percent	percent						
25	560	±20	7	28	76	15	28.8	24	-72	+20	+25	T3	0037	0125	0213	0301	0389
25	560	±10	7	28	76	15	28.8	24	-72	+20	+25	T3	0038	0126	0214	0302	0390
25	680	±20	8	32	63	15	28.8	19	-72	+25	+30	T4	0039	0127	0215	0303	0391
25	680	±10	8	32	63	15	28.8	19	-72	+25	+30	T4	0040	0128	0216	0304	0392
30	56	±20	2	9	22	20	34.5	100	-38	+12	+15	T1	0041	0129	0217	0305	0393
30	56	±10	2	9	22	20	34.5	100	-38	+12	+15	T1	0042	0130	0218	0306	0394
30	220	±20	3	16	42	20	34.5	36	-60	+13	+16	T2	0043	0131	0219	0307	0395
30	220	±10	3	16	42	20	34.5	36	-60	+13	+16	T2	0044	0132	0220	0308	0396
30	470	±20	8	32	64	20	34.5	25	-65	+20	+25	T3	0045	0133	0221	0309	0397
30	470	±10	8	32	64	20	34.5	25	-65	+20	+25	T3	0046	0134	0222	0310	0398
30	560	±20	9	36	55	20	34.5	20	-65	+25	+30	T4	0047	0135	0223	0311	0399
30	560	±10	9	36	55	20	34.5	20	-65	+25	+30	T4	0048	0136	0224	0312	0400
50	33	±20	2	9	12.3	30	57.5	135	-29	+10	+12	T1	0049	0137	0225	0313	0401
50	33	±10	2	9	12.3	30	57.5	135	-29	+10	+12	T1	0050	0138	0226	0314	0402
50	120	±20	4	24	22.5	30	57.5	49	-42	+12	+15	T2	0051	0139	0227	0315	0403
50	120	±10	4	24	22.5	30	57.5	49	-42	+12	+15	T2	0052	0140	0228	0316	0404
50	270	±20	8	32	37	30	57.5	29	-46	+20	+25	T3	0053	0141	0229	0317	0405
50	270	±10	8	32	37	30	57.5	29	-46	+20	+25	T3	0054	0142	0230	0318	0406
50	330	±20	9	36	38	30	57.5	22	-46	+25	+30	T4	0055	0143	0231	0319	0407
50	330	±10	9	36	38	30	57.5	22	-46	+25	+30	T4	0056	0144	0232	0320	0408
60	27	±20	3	12	10.2	40	69.0	144	-24	+10	+12	T1	0057	0145	0233	0321	0409
60	27	±10	3	12	10.2	40	69.0	144	-24	+10	+12	T1	0058	0146	0234	0322	0410
60	100	±20	4	20	19	40	69.0	54	-36	+12	+15	T2	0059	0147	0235	0323	0411
60	100	±10	4	20	19	40	69.0	54	-36	+12	+15	T2	0060	0148	0236	0324	0412
60	220	±20	8	32	30	40	69.0	29	-40	+16	+20	T3	0061	0149	0237	0325	0413
60	220	±10	8	32	30	40	69.0	29	-40	+16	+20	T3	0062	0150	0238	0326	0414
60	270	±20	9	36	27	40	69.0	23	-45	+20	+25	T4	0063	0151	0239	0327	0415
60	270	±10	9	36	27	40	69.0	23	-45	+20	+25	T4	0064	0152	0240	0328	0416
75	22	±20	3	12	8.5	50	86.2	157	-19	+10	+12	T1	0065	0153	0241	0329	0417
75	22	±10	3	12	8.5	50	86.2	157	-19	+10	+12	T1	0066	0154	0242	0330	0418
75	82	±20	4	24	15.2	50	86.2	63	-30	+12	+15	T2	0067	0155	0243	0331	0419
75	82	±10	4	24	15.2	50	86.2	63	-30	+12	+15	T2	0068	0156	0244	0332	0420
75	180	±20	9	36	24.4	50	86.2	30	-35	+16	+20	T3	0069	0157	0245	0333	0421
75	180	±10	9	36	24.4	50	86.2	30	-35	+16	+20	T3	0070	0158	0246	0334	0422
75	220	±20	10	40	37.0	50	86.2	24	-40	+20	+25	T4	0071	0159	0247	0335	0423
75	220	±10	10	40	37.0	50	86.2	24	-40	+20	+25	T4	0072	0160	0248	0336	0424

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TABLE I. Capacitor characteristics - Continued.

Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at			Case size	PIN M39006/21- Failure rate level (%/1,000 hour)				
			+25°C	+85°C & +125°C					-55°C	+85°C	+125°C		L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
volts, dc	µF	percent	µA	µA	percent	volts, dc	volts, dc	ohms	percent	percent	percent						
100	10	±20	3	12	4.5	65	115	200	-17	+10	+12	T1	0073	0161	0249	0337	0425
100	10	±10	3	12	4.5	65	115	200	-17	+10	+12	T1	0074	0162	0250	0338	0426
100	39	±20	5	24	10.4	65	115	80	-20	+12	+15	T2	0075	0163	0251	0339	0427
100	39	±10	5	24	10.4	65	115	80	-20	+12	+15	T2	0076	0164	0252	0340	0428
100	68	±20	10	40	11.3	65	115	40	-30	+14	+16	T3	0077	0165	0253	0341	0429
100	68	±10	10	40	11.3	65	115	40	-30	+14	+16	T3	0078	0166	0254	0342	0430
100	120	±20	12	48	25	65	115	30	-35	+15	+17	T4	0079	0167	0255	0343	0431
100	120	±10	12	48	25	65	115	30	-35	+15	+17	T4	0080	0168	0256	0344	0432
125	6.8	±20	3	12	6.0	85	144	300	-14	+10	+12	T1	0081	0169	0257	0345	0433
125	6.8	±10	3	12	6.0	85	144	300	-14	+10	+12	T1	0082	0170	0258	0346	0434
125	27	±20	5	24	7.2	85	144	90	-18	+12	+15	T2	0083	0171	0259	0347	0435
125	27	±10	5	24	7.2	85	144	90	-18	+12	+15	T2	0084	0172	0260	0348	0436
125	47	±20	10	40	7.9	85	144	50	-26	+14	+16	T3	0085	0173	0261	0349	0437
125	47	±10	10	40	7.9	85	144	50	-26	+14	+16	T3	0086	0174	0262	0350	0438
125	82	±20	12	48	17.4	85	144	32	-30	+15	+17	T4	0087	0175	0263	0351	0439
125	82	±10	12	48	17.4	85	144	32	-30	+15	+17	T4	0088	0176	0264	0352	0440

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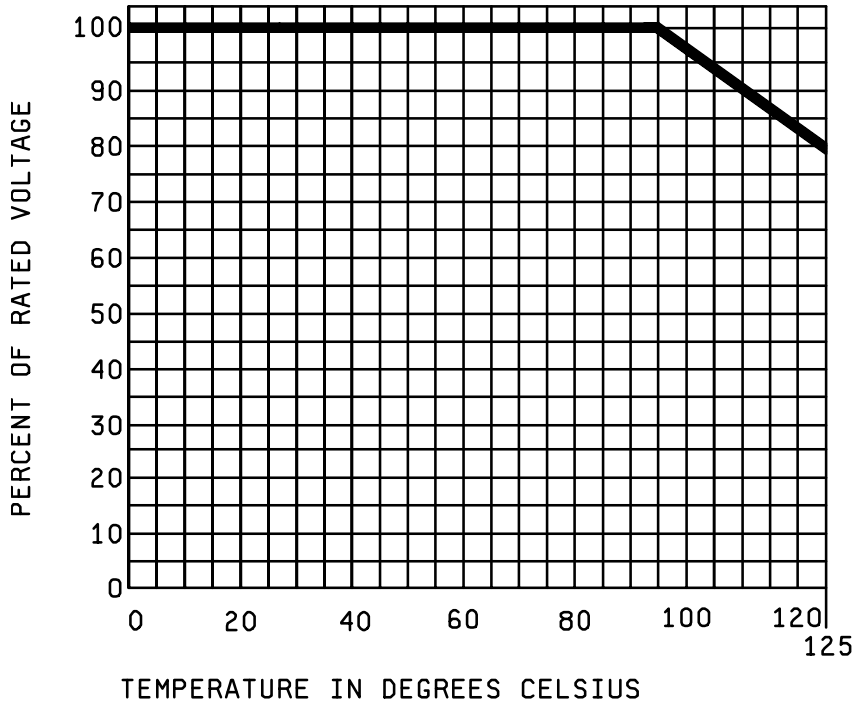


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:

Dimensions and configuration - See figure 1.

Case type: Tubular, insulated.

Seal type - Hermetic (glass-to-tantalum anode).

Terminals - Axial-wire lead:

Anode (type N51 of MIL-STD-1276).

Cathode (type C51 of MIL-STD-1276).

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

Rated voltage: See table I.

Capacitance (Cap.) (nom): See table I.

Cap. tolerance: See table I.

FR level: See table I.

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Constant voltage conditioning: As specified in MIL-PRF-39006.

DC leakage (DCL) (max) (at 25°C): See table I.

Cap. (nom): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (max): See table I.

Seal: Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G).

Thermal shock: Method 107 of MIL-STD-202, condition A (with step 3 at +125°C). Number of cycles = 30.

DCL: 125 percent (max) of 25°C value of table I.

ΔCap.: Within ±5 percent of initial measurement.

DF: 115 percent (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:

Pull: Method 211 of MIL-STD-202, condition A.

Wire-lead bend: In accordance with MIL-PRF-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.

DCL: 125 percent (max) of 25°C value of table I.

ΔCap.: Within ±8 percent of initial measurement.

DF: 115 percent (max) of initial requirement.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202. 2,000 V dc, minimum.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 100, megohms, minimum.

Low temperature (storage): Method 502 of MIL-STD-810.

DCL: See table I.

ΔCap.: Within ±5 percent of initial measurement.

DF: See table I.

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Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.

ΔCap.: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

Imp. See table I.

ΔCap.: See table I.

Step 3 (+25°C):

DCL: See table I.

ΔCap.: Within ±5 percent of step 1 value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

ΔCap.: See table I.

DF: See table I.

Step 5 (+125°C):

DCL: See table I.

ΔCap.: See table I.

DF: See table I.

Step 6 (+25°C):

DCL: See table I.

ΔCap.: Within ±5 percent of step 1 value.

DF: See table I.

Reverse voltage: Not applicable.

Life at 85°C: Method 108 of MIL-STD-202.

2,000 hours (qualification):

DCL: See table I.

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ΔCap.: Within +10 percent to -25 percent of initial measurement for 6-volt units;
within +10 percent to -20 percent of initial measurement for 8- and 10- volt units;
within +10 percent to -15 percent of initial measurement for 15-volt units; and
within ± 10 percent of initial measurement for all other units.

DF: See table I.

10,000 hours (group B) and extended life:

DCL (85°C): 125 percent (max) of table I value.

DCL (25°C): See table I.

ΔCap.: Within +10 percent, -60 percent of initial measurement.

DF: 200 percent (max) of table I value.

DWV: 2,000 V dc, minimum.

IR.: 100 megohms, minimum.

Vacuum life:

DCL: See table I.

ΔCap.: Within ±20 percent of initial measurement.

DF: See table I.

Weight loss: 4.1 mg, maximum.

Life at 125°C: Method 108 of MIL-STD-202.

DCL: See table I.

ΔCap.: Within +10 percent to -25 percent of initial measurement for 6-volt units;
within +10 percent to -20 percent of initial measurement for 8- and 10- volt units;
within +10 percent to -15 percent of initial measurement for 15-volt units; and
within ± 10 percent of initial measurement for all other units.

DF: See table I.

DWV: 2,000 V dc, minimum.

IR.: 100 megohms, minimum.

10,000 hours (group B):

DCL (125°C): 125 percent (max) of table I value.

DCL (25°C): See table I.

ΔCap.: Within +10 percent, -60 percent of initial measurement.

DF: 200 percent (max) of table I value.

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DWV: 2,000 V dc, minimum.

IR.: 100 megohms, minimum.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 feet).

Test voltage: Rated voltage.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

DCL: See table I.

ΔCap.: Within ±5 percent of initial measurement.

DF: See table I.

Marking: In accordance with MIL-PRF-39006.

Polarity: Series of plus (+) symbols encircling positive end (see figure 1).

Voltage groups (85°C rated voltage):

I	6 to 15 V inclusive
II	25 to 75 V inclusive.
III	100 V and 125 V.

Custodians:
Army - CR
Navy - EC
Air Force - 11

Review activities:
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

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